Chapter 1

Demo problem: Flow in a 2D channel with an oscillating wall revisited – AlgebraicElements: How to customise the mesh update

In many previous examples we illustrated how `oomph-lib's Domain` and `MacroElement` objects facilitate the generation of refineable (and non-refineable) meshes in domains with moving, curvilinear boundaries. Consult, for instance:

- The tutorial `"How to create refineable meshes in domains with curvilinear and/or moving boundaries"` for a general discussion of the methodology.

- The tutorial `"Spatially adaptive solution of the 2D unsteady heat equation with flux boundary conditions in a moving domain: ALE methods"` for an example involving the unsteady heat equation.

- The tutorials

  - `"Finite-Reynolds-number flow inside an oscillating ellipse"`

  or

  - `"Flow in a 2D channel with an oscillating wall"`

  for examples involving the Navier-Stokes equations.

The two main features of the `MacroElement/Domain` - based node-update are that, once the curvilinear domain is represented by a `Domain` object

1. Any subsequent mesh refinement will respect the curvilinear domain boundaries.

2. The udate of the nodal positions in response to a change in the position of the domain's curvilinear boundaries may be performed by the "black-box" function `Mesh::node_update()`.

The availability of a "black-box" node-update procedure is very convenient but in some applications it may be desirable (or even necessary) to provide a customised node-update function, either because the mesh deformation generated by the "black-box" procedure is not appropriate or because it is not efficient. The latter problem arises particularly in fluid-structure interaction problems.

In this tutorial we shall demonstrate an alternative node-update technique, based on `oomph-lib's AlgebraicNode`, `AlgebraicElement`, and `AlgebraicMesh` classes. The key feature of this approach is that it allows "each node to update its own position". This is in contrast to the `Domain/MacroElement-based` approach in which we can only update the nodal positions of *all* nodes in the mesh simultaneously – not a particularly sparse operation! [We note that the `Node` and `FiniteElement` base classes provide the virtual functions `Node::node_update()` and `FiniteElement::node_update()`. These functions are intended to be used for node-by-node or element-by-element node-updates but in their default implementations they are empty. Hence no node-update is performed unless these functions are overloaded in derived classes such as `AlgebraicNode` and `AlgebraicElement`.]

1.1 Overview

The idea behind the algebraic node-updates is simple: The `AlgebraicMesh` class (the base class for meshes containing `AlgebraicElements` and `AlgebraicNodes`) contains the pure virtual function

```
void AlgebraicMesh::algebraic_node_update(..., AlgebraicNode* node_pt)=0;
```

which must be implemented in every specific `AlgebraicMesh`. Its task is to update the position of the node specified by the pointer-valued argument.

The specific implementation of the node-update operation is obviously problem-dependent but it is easy to illustrate the general procedure by considering the collapsible channel mesh sketched below:

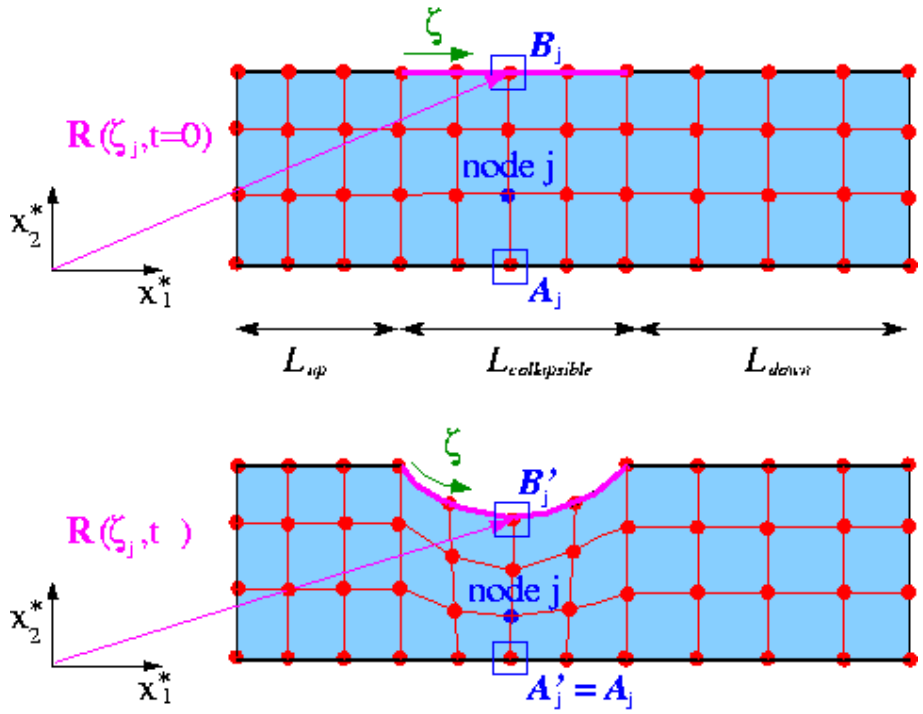


Figure 1.1 Sketch of the algebraic node-update procedure for the collapsible channel mesh.

The upper figure shows the mesh in the undeformed domain in which wall shape is parametrised by the intrinsic (Lagrangian) coordinate ζ as $\mathbf{R}(\zeta, t=0) = (L_{up} + \zeta, 1)^T$. In this configuration each node is located at a certain fraction along the vertical lines across the channel. For instance, the j-th node (drawn in blue) is located at a fraction of $\omega_j = 1/3$ along the straight line that connects reference point $\mathbf{A}_j$ on the bottom wall to reference point $\mathbf{B}_j$ on the flexible upper wall. We note that reference point $\mathbf{B}_j$ may be identified by its Lagrangian coordinate ζ_j on the wall.
The lower figure shows a sketch of the deformed domain and illustrates a possible algebraic node-update strategy: Given the new wall shape, described by $\mathbf{R}(\zeta, t)$, we position each node on the straight line that connects its reference point $\mathbf{A'}_j = \mathbf{A}_j$ on the lower wall to the reference point $\mathbf{B'}_j = \mathbf{R}(\zeta_j, t)$ on the deformable upper wall.
To perform this node-update procedure for a specific node, we generally have to store

- A pointer to one (or more) `GeomObject(s)` that define the curvilinear domain boundaries.
- A number of node-specific reference values (such as ω_j, ζ_j, and the x_1 -coordinate of point $\mathbf{A}_j$, $X_j^{(A)}$, say) that establish the node's position relative to these `GeomObjects`.
- A pointer to the `AlgebraicMesh` that implements the node-update procedure.

Since the node-update data is node-specific, we provide storage for it in the `AlgebraicNode` class – a class that is derived from the `Node` class. The node-update function itself is shared by many nodes in the mesh and is implemented in `AlgebraicMesh::algebraic_node_update(...)`. This function extracts the node-update parameters, ω_j, ζ_j, and $X_j^{(A)}$ and the pointer to the `GeomObject` that parametrises the wall, `wall_pt`, say, from the `AlgebraicNode` passed to it. With these parameters, the position of reference point $\mathbf{A'}_j$ is given by $\mathbf{A'}_j = (X_j^{(A)}, 0)^T$ while the coordinates of reference point $\mathbf{B'}_j = \mathbf{R}(\zeta_j, t)$ may be obtained from a call to `wall_pt->position(...)`. The nodal position $\mathbf{r}_j$ may then be updated via

$$\mathbf{r}_j = \mathbf{A}_j' + \omega_j \left(\mathbf{B}_j' - \mathbf{A}_j'\right)$$

1.2 How to use existing AlgebraicMeshes

To use the algebraic node-update capabilities of an existing `AlgebraicMesh`, all `Nodes` must be replaced by `AlgebraicNodes` to allow the storage of the node-specific node-update parameters. Recall that within `oomph-lib Nodes` are usually created by the elements, using the function `FiniteElement::construct_node(...)` whose argument specifies the local node number of the newly created `Node` within the element that creates it. (The specification of the local node number is required to allow the `FiniteElement::construct_node(...)` function to create a `Node` with the appropriate number of values and history values. For instance, in 2D Taylor-Hood Navier-Stokes elements, the elements' vertex nodes have to store three values, representing the two velocity components and the pressure, whereas all other nodes only require storage for the two velocity components.)
To ensure that the `FiniteElement::construct_node(...)` function creates `AlgebraicNodes` rather than "ordinary" `Nodes`, we provide the templated wrapper class

```
template<ELEMENT>
class AlgebraicElement<ELEMENT> : public ELEMENT
```

which overloads the `FiniteElement::construct_node(...)` function so that it creates `AlgebraicNodes` instead. In most other respects, the "wrapped" element behaves exactly as the underlying `ELEMENT` itself. To use an existing `AlgebraicMesh` with a certain element type, `QTaylorHoodElement<2>`, say, we simply "upgrade" the element to an `AlgebraicElement` by specifying the element type as `AlgebraicElement<QTaylorHoodElement<2> >`.

The changes to an existing driver code that performs the node update by the default `Domain/MacroElement` methodology are therefore completely trivial. You may wish to compare the driver code `collapsible_channel.cc`, discussed in an earlier example, to the driver code `collapsible_channel_algebraic.cc` in which the fluid domain is discretised by the `MyAlgebraicCollapsibleChannelMesh`, discussed below.

1.3 How to create a new AlgebraicMesh – a basic example

To illustrate how to create a new `AlgebraicMesh`, we will now discuss the implementation of the `MyAlgebraicCollapsibleChannelMesh` – an `AlgebraicMesh` - version of the `CollapsibleChannelMesh`, used in the earlier example.

1.3.1 The class definition

We construct the mesh by multiple inheritance from the `AlgebraicMesh` base class, and the already-existing `CollapsibleChannelMesh`. The constructor first calls the constructor of the underlying `CollapsibleChannelMesh` (thus "recycling" the basic mesh generation process, i.e. the generation of nodes and elements etc.) and then adds the algebraic node-update information by calling the function `setup_algebraic_node_update()`.

```
//==========start_algebraic_mesh=================================
/// Collapsible channel mesh with algebraic node update
//================================================================
template<class ELEMENT>
class MyAlgebraicCollapsibleChannelMesh :
  public AlgebraicMesh,
  public virtual CollapsibleChannelMesh<ELEMENT>
{

public:

 /// Constructor: Pass number of elements in upstream/collapsible/
 /// downstream segment and across the channel; lengths of upstream/
 /// collapsible/downstream segments and width of channel, pointer to
 /// GeomObject that defines the collapsible segment, and pointer to
 /// TimeStepper (defaults to the default timestepper, Steady).
 MyAlgebraicCollapsibleChannelMesh(const unsigned& nup,
                                   const unsigned& ncollapsible,
                                   const unsigned& ndown,
                                   const unsigned& ny,
                                   const double& lup,
                                   const double& lcollapsible,
                                   const double& ldown,
                                   const double& ly,
                                   GeomObject* wall_pt,
                                   TimeStepper* time_stepper_pt=
                                   &Mesh::Default_TimeStepper) :
  CollapsibleChannelMesh<ELEMENT>(nup, ncollapsible, ndown, ny,
                                  lup, lcollapsible, ldown, ly,
                                  wall_pt,
                                  time_stepper_pt)
  {
   // Setup algebraic node update operations
   setup_algebraic_node_update();
  }
```

We declare the interface for the pure virtual function `algebraic_node_update(...)`, to be discussed below, and implement a second pure virtual function, `AlgebraicMesh::update_node_update()`, that may be used to update reference values following a mesh adaptation. Since the current mesh is not refineable, we leave this function empty and refer to another example for a more detailed discussion of its role.

```
 /// Update nodal position at time level t (t=0: present;
 /// t>0: previous)
 void algebraic_node_update(const unsigned& t, AlgebraicNode*& node_pt);

 /// Update the geometric references that are used
 /// to update node after mesh adaptation.
 /// Empty -- no update of node update required
 void update_node_update(AlgebraicNode*& node_pt) {}
```

The protected member function `setup_algebraic_node_update()` will be discussed below.

```
protected:

 /// Function to setup the algebraic node update
 void setup_algebraic_node_update();


 /// Dummy function pointer
 CollapsibleChannelDomain::BLSquashFctPt Dummy_fct_pt;

};
```

1.3.2 Setting up the algebraic node update

When the function `setup_algebraic_node_update()` is called, the constructor of the underlying `CollapsibleChannelMesh` will already have created the mesh's elements and nodes, and the nodes will be located at their initial positions in the undeformed domain.

To set up the algebraic node-update data, we start by extracting the lengths of the upstream rigid section, L_{up}, and the length of the collapsible segment, $L_{collapsible}$, from the `CollapsibleChannelDomain:`

```
//============start_setup_algebraic_node_update=====================
/// Setup algebraic mesh update -- assumes that mesh has
/// initially been set up with the wall in its undeformed position.
//==================================================================
template<class ELEMENT>
void MyAlgebraicCollapsibleChannelMesh<ELEMENT>::setup_algebraic_node_update()
{

 // Extract some reference lengths from the CollapsibleChannelDomain.
 double l_up=this->domain_pt()->l_up();
 double l_collapsible=this->domain_pt()->l_collapsible();
```

Next, we loop over all `AlgebraicNodes` in the mesh and determine their current positions:

```
 // Loop over all nodes in mesh
 unsigned nnod=this->nnode();
 for (unsigned j=0;j<nnod;j++)
  {
   // Get pointer to node
   AlgebraicNode* nod_pt=node_pt(j);

   // Get coordinates
   double x=nod_pt->x(0);
   double y=nod_pt->x(1);
```

If the `j` -th node is located in the "collapsible" section of the mesh we determine its reference coordinate, ζ_j, on the wall and determine the coordinates of its reference point $\mathbf{B}_j$ from the `GeomObject` that represents the moving wall.

```
   // Check if the node is in the collapsible part:
   if ( (x>=l_up) && (x<=(l_up+l_collapsible)) )
    {

     // Get zeta coordinate on the undeformed wall
     Vector<double> zeta(1);
     zeta[0]=x-l_up;

     // Get position vector to wall:
     Vector<double> r_wall(2);
     this->Wall_pt->position(zeta,r_wall);
```

Just to be on the safe side, we check that the wall is actually in its undeformed configuration, as assumed.

```
     // Sanity check: Confirm that the wall is in its undeformed position
#ifdef PARANOID
     if ((std::abs(r_wall[0]-x)>1.0e-15)&&(std::abs(r_wall[1]-y)>1.0e-15))
      {
       std::ostringstream error_stream;
       error_stream
        « "Wall must be in its undeformed position when\n"
        « "algebraic node update information is set up!\n "
        « "x-discrepancy: " « std::abs(r_wall[0]-x) « std::endl
        « "y-discrepancy: " « std::abs(r_wall[1]-y) « std::endl;

       throw OomphLibError(
        error_stream.str(),
        OOMPH_CURRENT_FUNCTION,
        OOMPH_EXCEPTION_LOCATION);
      }
#endif
```

Next, we package the data required for the node-update operations (the pointers to `GeomObject(s)` and the reference values) into vectors. In the present example, the node-update operation only involves a single `Geom↩ Object:`

```
     // Only a single geometric object is involved in the node update operation
     Vector<GeomObject*> geom_object_pt(1);

     // The wall geometric object
     geom_object_pt[0]=this->Wall_pt;
```

We have three reference values: The x_1 - coordinate of point $\mathbf{A}_j$,

```
     // The update function requires three parameters:
     Vector<double> ref_value(3);

     // First reference value: Original x-position on the lower wall
     ref_value[0]=r_wall[0];
```

as well as the fractional height, ω_j,

```
     // Second reference value: Fractional position along
     // straight line from the bottom (at the original x-position)
     // to the point on the wall
     ref_value[1]=y/r_wall[1];
```

and the reference coordinate, ζ_j, along the wall:

```
     // Third reference value: Zeta coordinate on wall
     ref_value[2]=zeta[0];
```

The vectors of reference values and geometric objects are then passed to the node, together with the pointer to the

mesh (`this`) that implements the node-update function.

```
     // Setup algebraic update for node: Pass update information
     // to AlgebraicNode:
     nod_pt->add_node_update_info(
      this,               // mesh
      geom_object_pt,     // vector of geom objects
      ref_value);         // vector of ref. values
    }

  }

} //end of setup_algebraic_node_update
```

1.3.3 The node-update function

The function `algebraic_node_update(...)` reverses the setup process: It extracts the node-update data from the `AlgebraicNode` and updates its position at the `t` -th previous time-level:

```
//=============start_of_algebraic_node_update=====================
/// Perform algebraic mesh update at time level t (t=0: present;
/// t>0: previous)
//================================================================
template<class ELEMENT>
void MyAlgebraicCollapsibleChannelMesh<ELEMENT>::algebraic_node_update(
 const unsigned& t, AlgebraicNode*& node_pt)
{
```

We start by extracting the vectors of reference values and `GeomObjects` involved in this node's node-update, using the functions `AlgebraicNode::vector_ref_value()` which returns a Vector of reference values, and the function `AlgebraicNode::vector_geom_object_pt()` which returns a Vector of pointers to `GeomObjects`.

```
 // Extract reference values for update by copy construction
 Vector<double> ref_value(node_pt->vector_ref_value());

 // Extract geometric objects for update by copy construction
 Vector<GeomObject*> geom_object_pt(node_pt->vector_geom_object_pt());
```

Next, we translate these into local variables,

```
 // First reference value: Original x-position
 double x_bottom=ref_value[0];
 // Second reference value: Fractional position along
 // straight line from the bottom (at the original x-position)
 // to the point on the wall
 double fract=ref_value[1];
 // Third reference value: Zeta coordinate on wall
 Vector<double> zeta(1);
 zeta[0]=ref_value[2];

 // Pointer to wall geom object
 GeomObject* wall_pt=geom_object_pt[0];
```

and obtain the current wall position from the wall `GeomObject`.

```
 // Get position vector to wall at previous timestep t
 Vector<double> r_wall(2);
 wall_pt->position(t,zeta,r_wall);
```

Finally, we update the nodal position:

```
 // Assign new nodal coordinates
 node_pt->x(t,0)=x_bottom+fract*(r_wall[0]-x_bottom);
 node_pt->x(t,1)=         fract* r_wall[1];

}
```

Done!

1.4 Comments and Exercises

1.4.1 Comments

- **Element-by-element and global node updates:**

  As explained above, the re-implementation of the (empty) `Node::node_update()` function by `AlgebraicNode::node_update()` allows each node to "update its own position". The `AlgebraicMesh` and `AlgebraicElement` classes provide their own re-implementation of the `node_update()` functions in the `Mesh` and `FiniteElement` classes, and perform the node-updates by executing the `AlgebraicNode::node_update()` function of their constituent nodes.

- **Default node-update function:**

  We stress that, in the above example, it was only necessary to set up the node-update data for the nodes in the central "collapsible" part of the mesh as they are the only modes whose position is affected by the motion of the curvilinear boundary. This is possible because the `AlgebraicNode` constructor provides default assignments for the node-update data. In particular, the pointer to the Mesh that performs the node update is initialised by a pointer to the (static instantiation of a) `DummyMesh` whose `AlgebraicMesh::algebraic_node_update(...)` and `AlgebraicMesh::update_node_update(...)` functions are empty. This ensures that nodes for which these default assignments are not overwritten stay at their original position when the node udate is performed. This implementation provides a sensible default behaviour for the node update.

- **Multiple node-update functions:**

  In the simple example considered here, a single node-update function was sufficient to update the positions of all moving nodes in the mesh. In more complicated meshes, it may be be necessary to provide different node-update functions for nodes that are located in different parts of the mesh. To facilitate the implementation of such cases, it is possible to specify an identifier for the node-update function when calling the `AlgebraicNode::add_node_update_info(...)` function, using its alternative interface

```
void AlgebraicNode::add_node_update_info(
    const int& id,                                // ID of the node-update fct
    AlgebraicMesh* mesh_pt,                       // pointer to mesh
    const Vector<GeomObject*>& geom_object_pt,    // vector of geom objects
    const Vector<double>& ref_value);             // vector of ref. values
```

  When implementing the `AlgebraicMesh::algebraic_node_update(...)` function for a mesh that contains multiple node-update functions, the ID of the node-update function associated with a particular node can be obtained from `AlgebraicNode::node_update_fct_id()`, allowing the node-update function to take the appropriate action. (As an example, consider the implementation of the algebraic node-update function for the `RefineableAlgebraicFishMesh` in which different node-update functions are used to update the position of nodes in the fish's "body" and in its "tail".) If the `AlgebraicNode::add_node_update_info(...)` function is called without specifying an ID, a default ID of 0 is assigned.

- **Consistency between multiple node-update functions:**

  If a mesh contains multiple node-update functions, it is likely that some nodes are located at the interface between regions that are updated by different node-update functions. As indicated by the `"add_..."` in the `AlgebraicNode::add_node_update_info(...)` function, `AlgebraicNodes` may be associated with multiple node-update functions, though the different node-update functions associated with a node must, of course, give the same result. This may be verified by calling the `AlgebraicNode::self_test()` function for all nodes in an `AlgebraicMesh`.

  Note: In refineable `AlgebraicMeshes` (discussed in more detail below), `AlgebraicNodes` **must** be associated with all possible node-update functions to ensure that the reference values for newly created nodes are determined correctly during the adaptive mesh refinement.

- **Adaptivity:**

  Refineable `AlgebraicMeshes` may be created by multiple inheritance from a suitable `RefineableMesh` base class (e.g. the `RefineableQuadMesh` class), and setting up the required tree representation of the coarse initial mesh, exactly as for "ordinary" meshes (see the tutorial `"How to create simple refineable meshes"` for details). As an example, here is the class definition for a refineable version of the `MyAlgebraicCollapsibleChannelMesh` discussed above:

```
//===========start_refineable_algebraic_collapsible_channel_mesh======
```

```
/// Refineable version of the CollapsibleChannel mesh with
/// algebraic node update.
//======================================================================
template<class ELEMENT>
class MyRefineableAlgebraicCollapsibleChannelMesh :
  public RefineableQuadMesh<ELEMENT>,
  public virtual MyAlgebraicCollapsibleChannelMesh<ELEMENT>
{

public:


 /// Constructor: Pass number of elements in upstream/collapsible/
 /// downstream segment and across the channel; lengths of upstream/
 /// collapsible/downstream segments and width of channel, pointer to
 /// GeomObject that defines the collapsible segment, function pointer
 /// to "boundary layer squash function", and pointer to
 /// TimeStepper (defaults to the default timestepper, Steady).
 MyRefineableAlgebraicCollapsibleChannelMesh(const unsigned& nup,
                                             const unsigned& ncollapsible,
                                             const unsigned& ndown,
                                             const unsigned& ny,
                                             const double& lup,
                                             const double& lcollapsible,
                                             const double& ldown,
                                             const double& ly,
                                             GeomObject* wall_pt,
                                             TimeStepper* time_stepper_pt=
                                             &Mesh::Default_TimeStepper) :
  CollapsibleChannelMesh<ELEMENT>(nup, ncollapsible, ndown, ny,
                                  lup, lcollapsible, ldown, ly,
                                  wall_pt,
                                  time_stepper_pt),
  MyAlgebraicCollapsibleChannelMesh<ELEMENT>(nup, ncollapsible, ndown, ny,
                                             lup, lcollapsible, ldown, ly,
                                             wall_pt,
                                             time_stepper_pt)
  {
   // Build quadtree forest
   this->setup_quadtree_forest();
  }
};
```

Note that none of the functions defined in the non-refineable version of this mesh have to be re-implemented. This raises the question of how the node-update data for any newly created nodes are determined when the mesh is refined. By default, the reference values for any newly created `AlgebraicNodes` are determined by interpolation from the node's "father element". Furthermore, it is assumed
that the same `GeomObjects` are involved in the node-update function for the newly created node.
This default behaviour is appropriate for the mesh considered here. In other cases, (e.g. in the corresponding `fluid-structure interaction problem`), some or all of the node-update data may have to be recomputed when the mesh is adapted. Such updates may be performed by the `AlgebraicMesh::update_node_update(...)` function which is executed automatically whenever a refineable `AlgebraicMesh` is adapted.

- **Node updates for hanging nodes:**

  Recall that refineable meshes may contain hanging nodes whose position is constrained by its "master nodes". When the `AlgebraicNode::node_update(...)` function is called for a hanging node, it first updates the position of its master nodes (using their own node-update functions) and then updates its own constrained position accordingly.

- **Automatic finite-differencing with respect to geometric Data:**

  Algebraic node updates are particularly useful in fluid-structure interaction problems since they allow the development of "sparse" node update procedures in which each (solid mechanics) degree of freedom only affects the position of a small number of nodes in the fluid mesh. We will discuss fluid-structure interaction problems in more detail `elsewhere` but stress already that one of the major complications in such problems (and, in fact, in any free-boundary problem) is the need to evaluate the so-called shape derivatives – the derivatives of the "bulk" (here the Navier-Stokes) equations with respect to the degrees of freedom

(here the nodal positions in the wall mesh) that affect the position of the nodes in the "bulk" mesh.

The algebraic node update procedures discussed above are sufficiently general to handle such interactions. The shape/position of the `GeomObjects` that are involved in an `AlgebraicNode's` node update, may depend on unknowns in the overall problem. Such unknowns constitute a `GeomObject's` geometric `Data` which may be obtained from its member function `GeomObject::geom_data_pt(...)`. `AlgebraicElements` use finite differencing to include the shape derivatives (i.e. the derivatives of the residuals of the underlying (wrapped) element with respect to the geometric `Data` involved in the element's node update) into the element's Jacobian matrix.

- **Additional sanity checks and other generalisations:**

The `MyAlgebraicCollapsibleChannelMesh` and its refineable equivalent are slightly simplified versions of the meshes in `oomph-lib's src/meshes` directory. These meshes contain a number of additional sanity checks that we omitted here for the sake of brevity. Furthermore, these meshes can be used with `GeomObjects` that comprise "sub-"`GeomObjects`, a feature that is essential in problems with proper fluid-structure interaction. We will discuss this in another example.

1.4.2 Exercises

1. If you inspect the source code for the `MyAlgebraicCollapsibleChannelMesh`, you will notice that the mesh has an additional constructor that allows the specification of the "boundary layer squash function" first introduced in the original `CollapsibleChannelMesh`. Explain why in the `AlgebraicMesh` version of this mesh, the function pointer to the "boundary layer squash function" can only be specified via the constructor – the access function is deliberately broken.

1.5 Source files for this tutorial

- The source files for this tutorial are located in the directory:

demo_drivers/navier_stokes/collapsible_channel/

- The driver code is:

demo_drivers/navier_stokes/collapsible_channel/collapsible_channel_algebraic.cc

---

1.6 PDF file

A pdf version of this document is available. \